

EXAM 7 – SPRING 2014

4. (3.75 points)

Given the following loss information:

<u>Cumulative Reported Losses (\$000)</u>						
<u>Year</u>	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>	<u>48 Months</u>	<u>60 Months</u>	<u>72 Months</u>
2008	200	1,000	1,500	1,800	2,100	2,100
2009	400	800	1,000	1,500	1,800	
2010	800	1,500	2,000	2,200		
2011	1,200	2,000	2,200			
2012	400	1,800				
2013	200					

<u>t-statistic for 0.90 at Various Degrees of Freedom (DF)</u>							
<u>DF</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>
t-statistic	3.08	1.89	1.64	1.53	1.48	1.44	1.42

- Assume $T = r [(n - 2) / (1 - r^2)]^{1/2}$ is t -distributed with $n - 2$ degrees of freedom where r is the sample correlation coefficient.

a. (3.25 points)

Evaluate whether the triangle above meets the assumption that the age-to-age factors are independent from 12 months to 36 months, using a 10% t -statistic significance standard.

b. (0.5 point)

State two other basic chain-ladder assumptions needed for least squares optimality.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 4	
TOTAL POINT VALUE: 3.75	LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as: • Chain ladder • Cape Cod • Chain ladder plus calendar-year effects • Bornhuetter-Ferguson
SAMPLE ANSWERS	
Part a: 3.25 points	
Sample Answer 1	
Use Pearson's Correlation: $r = \frac{E[XY] - E[X] \times E[Y]}{\sigma_X \times \sigma_Y}$	
<u>AY</u>	<u>12-24 LDF's</u> <u>24-36 LDF's</u>
08	5.000 1.500
09	2.000 1.250
10	1.875 1.333
11	1.667 1.100
<u>AY</u>	<u>X-Values</u> <u>Y-Values</u> (LDF's -1)
08	4.000 0.500
09	1.000 0.250
10	0.875 0.333
11	0.667 0.100
$\Sigma XY = 2.606$ $E[XY] = 2.606/4 = .651$	
$r = \frac{0.651 - 1.636 \times 0.295}{1.37 \times .144}$ $= 0.854$	
$T = 0.854 \times ((4 - 2)/(1 - .854^2))^{1/2} = 2.321$ Use D.F. = 4-2 = 2 → Compare to 1.89 from table 2.321 > 1.89, so assumption of independence from 12-36 months fails.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2

Use Pearson's Correlation:

$$r = \frac{\sum \{[X - E[X]] \times [Y - E[Y]]\}}{\sqrt{\sum [X - E[X]]^2} \sqrt{\sum [Y - E[Y]]^2}}$$

<u>AY</u>	<u>12-24 LDF's</u>	<u>24-36 LDF's</u>
08	5.000	1.500
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<u>AY</u>	<u>X-Values</u>	<u>Y-Values</u> (LDF's -1)
08	4.000	0.500
09	1.000	0.250
10	0.875	0.333
11	0.667	0.100

$$E[X] = 1.635$$

$$E[Y] = .2958$$

$$r = \frac{0.673}{2.74 \times .289}$$

$$= 0.849$$

$$T = r \times ((n - 2)/(1 - r^2))^{1/2}$$

$$T = 0.849 \times ((4 - 2)/(1 - .849^2))^{1/2} = 2.28$$

$$T_{4-2} = 1.89$$

2.28 > 1.89 – Using a 10% t statistic this assumption is not met.

Part b: 0.5 point

Sample Answer 1

Assumes losses in the next period are equal to a factor based on age times cumulative losses to date

Assumes variance of incremental losses is proportionate to cumulative losses and a factor based on age.

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2

This formula is from Venter the above tests that the accident year losses are independent.

Two other assumptions are:

$E\{q(w,d+1) | \text{data to } w + d\} = f(d)c(w,d)$ – next period's losses are losses reported to date times a factor that depends on age

$\text{Var}\{q(w,d+1) | \text{data to } w + d\} = a(d, c(w,d))$ – variance is proportional to losses reported to date and a factor that varies by age.

EXAMINER'S REPORT

Overall, many candidates performed very well on this question. Based on the knowledge statements candidates should know key assumptions of the chain ladder models and how to test these assumptions. The core of the question is determining whether the age-to-age factors are independent. This implies correlation and the candidate should know the formula for correlation as well as how to calculate it and the value of "n".

Part a

In order to receive full credit candidates needed to calculate link ratios at 12-24 months as well as 24-36 months and calculate the correlation coefficient based on the Pearson's correlation formula. Based on the correlation and the appropriate t statistic, candidates were expected to make the appropriate conclusion that the null hypothesis is rejected and age to age are correlated at the 10% level of significance.

Some candidates did not receive full credit for a variety of reasons:

Some candidates failed to calculate the degrees of freedom correctly (D.F. = n-2).

Other candidates chose a rank correlation approach. Although the idea is correct, rank correlation is appropriate for sets of data where there are no ties between the items being tested. These ties were present in the data being tested and therefore this approach is incorrect.

Other candidates incorrectly relied on the actual losses rather than link ratios.

Part b

Many candidates stated the independence of accident years which was the assumption being tested in part a. However the question asked for two other assumptions. Full credit was given to those candidates who understood that future expected incremental losses are proportional to cumulative losses to date and that the variance of future incremental losses are also proportional to cumulative losses to date.